EXAMINATION RESULT AND GRACE-MARK VALIDATION

Complete Python Program

```
# Examination Result and Grace-Mark Validation

num_subjects = int(input("Enter number of subjects: "))

marks = []
max_marks = 100
pass_mark = 35
grace_limit = 5

failed_subjects = 0
total = 0

for i in range(num_subjects):
    mark = int(input(f"Enter marks for Subject {i + 1}: "))

    if mark < 0 or mark > max_marks:
        print("Invalid marks entered!")
        exit()

    # Apply grace marks if eligible
    if mark < pass_mark and (pass_mark - mark) <= grace_limit:
        grace = pass_mark - mark
        mark += grace
        print(f"Grace Marks Applied: {grace} (New Marks = {mark})")

    marks.append(mark)
    total += mark

    if mark < pass_mark:
        failed_subjects += 1

percentage = (total / (num_subjects * max_marks)) * 100

print("\n----- RESULT -----")
print("Final Marks:", marks)
print("Total Marks:", total)
print(f"Percentage: {percentage:.2f}%")

if failed_subjects > 0:
    print("Result: FAIL")
else:
    if percentage >= 75:
        print("Classification: DISTINCTION")
    elif percentage >= 60:
        print("Classification: FIRST CLASS")
    elif percentage >= 50:
        print("Classification: SECOND CLASS")
    else:
        print("Classification: PASS")
```

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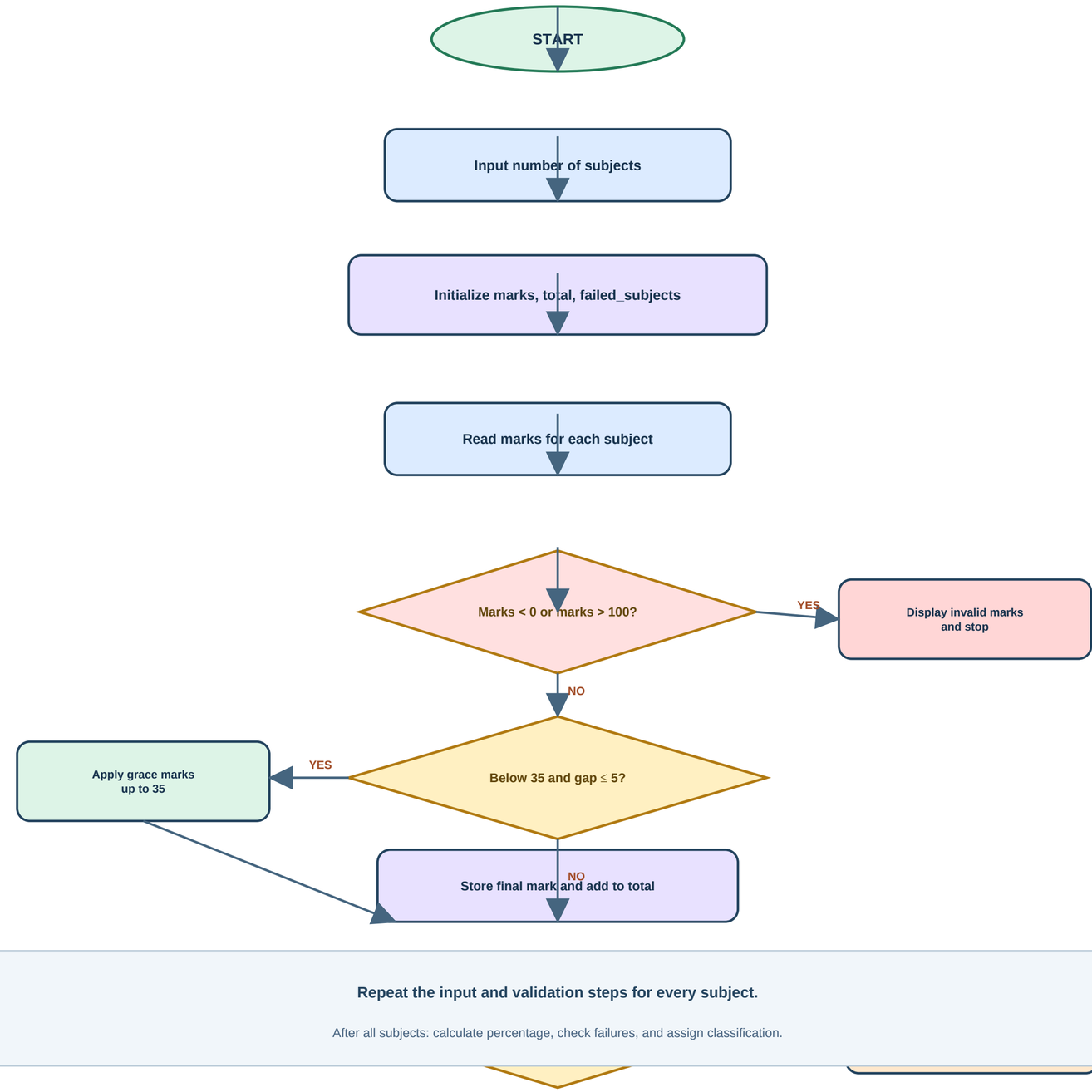
Colorful algorithm flowchart



Repeat the input and validation steps for every subject.

After all subjects: calculate percentage, check failures, and assign classification.

EXAMINATION RESULT AND GRACE-MARK VALIDATION

Assignment Matter • Algorithm • Pseudocode • Sample Output

1. Problem Statement

Develop a Python program to calculate a student's examination result. The program reads marks for a given number of subjects, checks that every mark is between 0 and 100, applies grace marks when a student is short of the pass mark by not more than 5 marks, calculates the total and percentage, counts failed subjects, and displays the final result with the appropriate classification.

2. Rules and Conditions

Item	Rule
Maximum marks	100 marks per subject
Pass mark	35 marks per subject
Grace-mark limit	A student who scores 30–34 receives only the marks needed to reach 35.
Invalid input	Marks below 0 or above 100 are rejected and the program stops.
Overall result	If one or more subjects remain below 35, the result is FAIL.
Classification	75% or above: Distinction; 60% or above: First Class; 50% or above: Second Class; otherwise Pass.

3. Algorithm

1. Start the program.

2. Read the number of subjects and initialize the marks list, total, and failed-subject count.

3. For each subject, read the mark and check whether it is within 0 to 100.

4. If the mark is invalid, display an error message and stop the program.

5. If the mark is below 35 and the difference from 35 is at most 5, add the required grace marks.

6. Store the final mark, add it to the total, and increase the failed-subject count if it is still below 35.

7. After processing all subjects, calculate the percentage.

8. If any subject has failed, display FAIL. Otherwise, display the classification based on the percentage.

9. Stop the program.

4. Pseudocode

```
BEGIN
    INPUT num_subjects
    SET max_marks = 100, pass_mark = 35, grace_limit = 5
    SET total = 0, failed_subjects = 0
    CREATE empty marks list

    FOR each subject from 1 to num_subjects
        INPUT mark
        IF mark < 0 OR mark > 100
            DISPLAY "Invalid marks entered"
            STOP
        END IF

        IF mark < pass_mark AND (pass_mark - mark) <= grace_limit
            SET grace = pass_mark - mark
            SET mark = mark + grace
            DISPLAY grace marks applied
        END IF

        ADD mark to marks
        total = total + mark
        IF mark < pass_mark
            failed_subjects = failed_subjects + 1
        END IF
    END FOR

    percentage = (total / (num_subjects * max_marks)) * 100
    DISPLAY marks, total, percentage

    IF failed_subjects > 0
        DISPLAY "FAIL"
    ELSE IF percentage >= 75
        DISPLAY "DISTINCTION"
    ELSE IF percentage >= 60
        DISPLAY "FIRST CLASS"
    ELSE IF percentage >= 50
        DISPLAY "SECOND CLASS"
    ELSE
        DISPLAY "PASS"
    END IF
```

```
END
```

5. Sample Input and Output

```
Enter number of subjects: 5
Enter marks for Subject 1: 82
Enter marks for Subject 2: 33
Grace Marks Applied: 2 (New Marks = 35)
Enter marks for Subject 3: 67
Enter marks for Subject 4: 74
Enter marks for Subject 5: 91

----- RESULT -----
Final Marks: [82, 35, 67, 74, 91]
Total Marks: 349
Percentage: 69.80%
Classification: FIRST CLASS
```

6. Concepts Used

Variables: num_subjects, marks, mark, total, failed_subjects, percentage. **Data types:** integer, float, and list. **Operators:** arithmetic (+, -, /, *), comparison (<, >, >=), and logical (and). **Conditional statements:** if, elif, else. **Loop:** for loop to process each subject. **List:** stores the final marks of all subjects.

7. Conclusion

This program validates examination marks, applies the specified grace-mark rule, calculates the total and percentage, and determines the student's result. It demonstrates the use of variables, lists, loops, operators, and conditional statements in Python.

Note: This solution follows the grace-mark rule given in the question. Actual grace-mark policies may differ between institutions.